

Study on the effect of slot width on two-elements wingsails

Thomas LEPERE,^{*} Giuliano ZENNARO,[†] and Hui DONG[‡]
STI - Mechanical Engineering, École Polytechnique Fédérale de Lausanne (EPFL)
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A 2D RANS parametric study was performed using Ansys Fluent to assess the role of the slot width between the two elements of an AC72 wingsail. The results highlight a complex relationship between angle of attack, camber and the effectiveness of the slot on the wingsail performances. Both an increase and decrease in the lift-to-drag ratio can be observed depending on the wingsail configuration. It is suggested that a larger gap be used if high angles of attack are expected, however the study is limited by its lack of proper stall modeling.

Keywords: AC72, wingsail, CFD, RANS, 2D, NACA, slot width, AoA, camber, lift, drag

NOMENCLATURE

α	Angle of attack	°
δ	Domain diameter	m
γ	Camber	°
D	Drag	N
L	Lift	N
s	Slot width	%
S_G	Global mesh size	m
V_A	Apparent wind speed	kn
V_S	Boat speed	kn
V_T	True wind speed	kn
z	Height	m
z_0	Roughness length	m
z_{ref}	Reference height	m

ACRONYMS

FVM: Finite Volume Method

LE: Leading Element

RANS: Reynolds-averaged Navier-Stokes

SST: Shear Stress Transport

TE: Trailing Element

I. INTRODUCTION

The America's Cup is the oldest international sailing competition, showcasing cutting-edge yacht design and technology. One important aspect of the competition is the fact that each team must design their own boat, meaning the winning teams are often the ones with the most technological development.

One of the prominent advancements in recent editions is the introduction of rigid sails called wingsails. These are inspired by aircraft wings and consist of two or more symmetric rigid wings which can be articulated to control angle of attack and camber. Although wingsails have existed for quite some time, the 2013 America's Cup AC72 has shown a leap in performance by combining the wingsail to the major reduction in drag brought by hydrofoils.

Much like airplane flaps, the gap between the two wingsail elements is of particular interest. This gap creates a leakage flow, where air flows from the high-pressure side of the leading element to the low-pressure side of the trailing element through the slot. This flow helps delay separation and promotes flow reattachment. The effectiveness of this mechanism is a complex relationship between the slot width, the angle of attack, and the relative camber of the wingsail elements [1].

The aim of this study is to gain understanding in the effect of the slot width on the wingsail performance under several camber and angle of attack configurations. To do so, a 2D Reynolds-averaged Navier-Stokes (RANS) numerical simulation was done with Ansys Fluent to solve the flow around the sail and calculate lift and drag.

II. WINGSAIL GEOMETRY

The wingsail was modeled as a simplified 2D geometry inspired from the 2013 America's Cup AC72. It comprises of two symmetrical NACA profiles: a NACA0024 Leading Element (LE), also referred to as the main element, and a NACA0012 Trailing Element (TE). For this 2D study, the dimensions of these elements are based on the maximum allowable sizes defined by the America's Cup regulations [2] for chord number 4 (LE of 3.213 m, TE of 6.141 m), as depicted in Figure 1. The TE, also re-

^{*} thomas.lepere@epfl.ch; SCIPER: 369279

[†] giuliano.zennaro@epfl.ch; SCIPER: 344437

[‡] hui.dong@epfl.ch; SCIPER: 377799

ferred to as the flap, can rotate about its hinge axis which is located at 90 % of the chord length of the LE, measured at chord number 0 (Fig. 1, and Fig. 2). This rotation angle is defined as camber (γ). The gap between the two elements is defined as the slot width, which is given as a percentage relative to the chord length of the LE, measured at chord number 0. The angle of attack (α) is defined as the angle between the incoming apparent wind direction and the line passing through the nose of the LE to the trailing edge of the TE. This line represents the effective chord line of the entire wingsail, accounting for both elements.

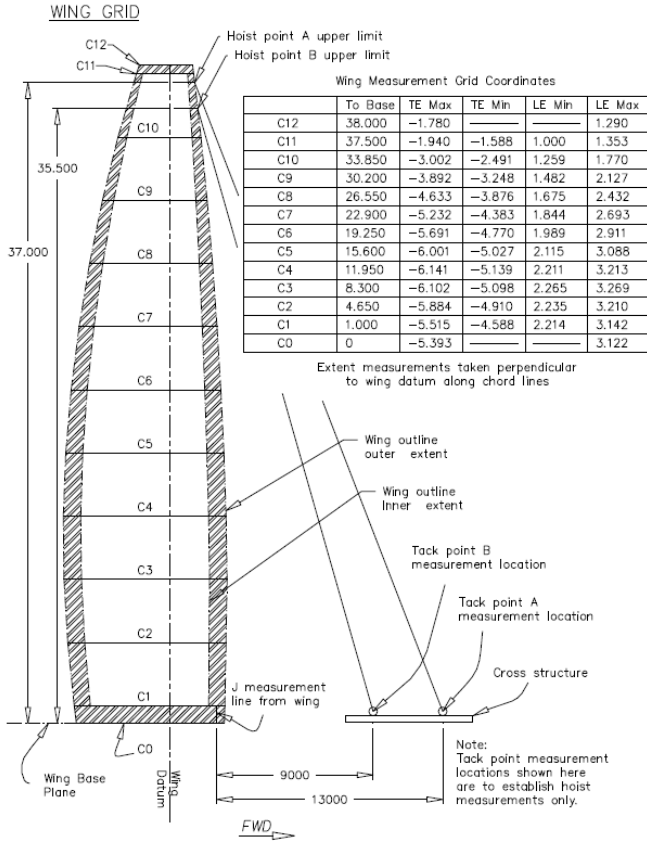


FIG. 1: AC72 wingsail main geometric rules from Ref. [2]. The rules define minimum and maximum allowable chord lengths for each element of the sail (LE and TE) and for 13 chords, numbered from the bottom to the top of the sail. Each chord number is assigned a distance to the base of the sail and all the dimensions are given in meters.

III. FLOW CONDITIONS

For this study, only one sailing condition was considered. The true wind speed was assumed to be 15 kn (7.7 ms^{-1}) with the boat sailing close hauled at $V_S = 21 \text{ kn}$ (10.8 ms^{-1}), 45° from the true wind. Although

the study is limited to 2D, the vertical wind gradient was used to calculate the actual wind speed at the selected height of the sail. The vertical wind speed gradient in the atmospheric boundary layer was modeled as a logarithmic law [3]:

$$V_T(z) = V_T(z_{ref}) \frac{\log z/z_0}{\log z_{ref}/z_0} \quad (1)$$

where $V_T(z_{ref})$ is the true wind speed at the reference height z_{ref} , here taken as 10 m, and z_0 is the roughness length which can be approximated as [3]:

$$z_0 \approx 5 \times 10^{-5} \frac{V_T(z_{ref})^2}{g} \quad (2)$$

Taking the boat speed and the vertical gradient into account, the apparent wind speed, $\vec{V}_A = \vec{V}_T - \vec{V}_S$, at chord number 4 was calculated as being equal to 33.89 kn (17.43 ms^{-1}). This speed was used as the windward boundary condition. The leeward boundary condition was simply enforcing gauge pressure. Although the speeds are reported in knots, all quantities were converted to the standard SI unit system for use in the simulations.

The fluid domain was modeled as air at 15°C and at atmospheric pressure. This yields a dynamic viscosity of $1.7894 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$ and a density of 1.225 kg m^{-3} . By using as a reference length the sum of the LE and TE chord lengths (9.4 m), the Reynolds number could be approximated as being equal to $Re \approx 1 \times 10^7$, which is a fully developed turbulent regime. Furthermore, the Mach number was evaluated at $M = 0.05$, which justified an incompressible simulation.

IV. GEOMETRIC PARAMETERS

To study multiple wingsail configurations, a parametric study was performed which enabled to easily vary angle of attack, camber and slot width. The ranges of these parameters that have been tested are summarized in Table I.

TABLE I: Ranges of geometric quantities varied for the wingsail simulations.

Angle of Attack α ($^\circ$)	Base Camber γ ($^\circ$)	Slot Width s (% of C0 LE)
0 - 15	15 - 35	0.5 - 5

To enable an accurate comparison to a complete 3D simulation which also models the vertical wind gradient, or to compare the results with those of another chord number while maintaining the same sail settings, the camber was adapted to account for twist. Because of the gradient in the true wind speed, the apparent wind

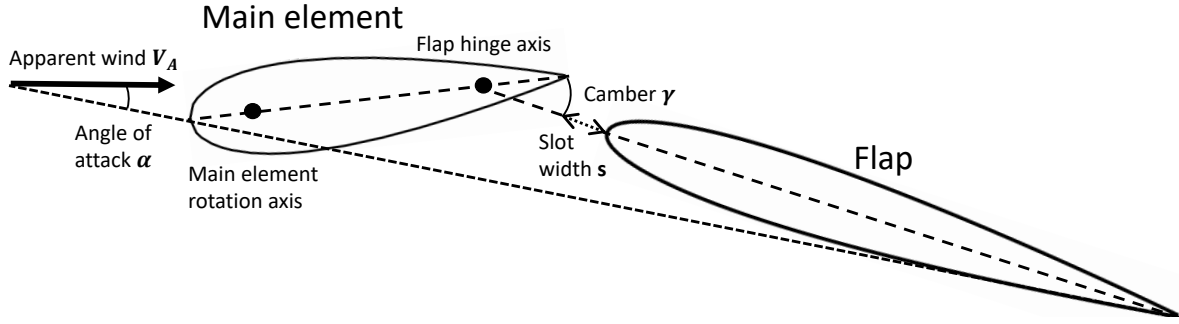


FIG. 2: Schematics of a chord of the AC72 wingsail and associated nomenclature.

not only has a gradient in magnitude, but also in direction. Thus, to keep a constant angle of attack throughout the height of the sail, it has to be twisted. For a wing-sail, this can be done by reducing camber with height, which is achieved by having multiple flap elements that are independently actuated (Fig. 3).

In this study, that means that the actual camber at chord number 4 is roughly 4° less than the base camber defined at the bottom of the sail. The actual camber naturally has to be recalculated for each geometric configuration, although the precise value does not change much. Unless otherwise specified, camber hereinafter refers to the base camber of the sail.

V. NUMERICAL METHOD

A. Type of Simulation and solver

A 2D RANS parametric study was conducted with Ansys Fluent, employing the Finite Volume Method (FVM) with quadratic elements for spatial discretization. The simulations were steady, incompressible and pressure-based, and were carried out using the $k-\omega$ Shear Stress Transport (SST) turbulence model. The equations are coupled with the least squares cell-based method for gradient computation, and second-order schemes are used for solving pressure, momentum, and turbulent kinetic energy.

The RANS equations, which represent the decomposition of the Navier-Stokes equations into mean and fluctuating components, were solved within Fluent. The velocity components are expressed as $u_i = \overline{u}_i + u'_i$, and scalar quantities, such as pressure, are expressed as $\varphi = \overline{\varphi} + \varphi'$, where the overline and prime represent mean and fluctuating components respectively. Substituting these expressions into the Navier-Stokes equations and omitting the overline for readability yields the following governing equations:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial X_i}(\rho u_i) = 0 \quad (3)$$

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial X_j}(\rho u_i u_j) = & \frac{\partial p}{\partial X_j} + \frac{\partial}{\partial X_j} \left[\mu \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right. \right. \\ & \left. \left. - \frac{2}{3} \delta_{ij} \frac{\partial u_l}{\partial X_l} \right) \right] + \frac{\partial}{\partial X_j}(-\rho \overline{u'_i u'_j}) \end{aligned} \quad (4)$$

The term $-\rho \overline{u'_i u'_j}$ is unknown, and is modeled using the turbulence model.

B. Solution options and computed quantities

A hybrid initialization was performed for 10 iterations. Drag (D) and lift (L) were computed and the solution was considered converged when the force ratio L/D converged within a tolerance of 0.001, or when 200 iterations were reached. The y^+ value was computed (see appendix) to ensure it met the turbulence model's requirements ($y^+ < 1$) and the mass imbalance (difference in flow rate between the inlet and outlet) was used to assess the quality of the simulations.

VI. MESH AND DOMAIN CONVERGENCE STUDY

A. Convergence criterion

The mesh and domain convergence analyses were conducted based on the lift-to-drag ratio. It was assumed that if the force ratio had converged, both lift and drag had also individually converged. This assumption's validity was manually verified, which highlighted that the ratio was the last quantity to converge. An incremental error of 3% and an absolute error (calculated as the difference between the current value and the value obtained with the finest mesh) of 3% were used as a threshold.

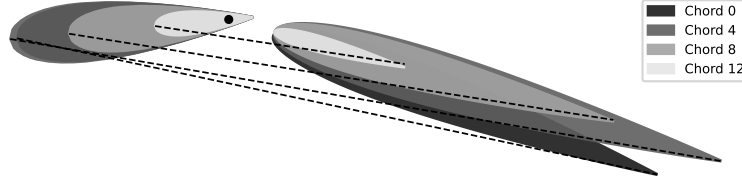


FIG. 3: Twist profile of the AC72 wingsail, illustrated using 4 chords shaded in varying tones of gray, corresponding to chord numbers 0, 4, 8, and 12. The twist profile is obtained by gradually reducing camber as a function of the height. The black dotted lines represent the effective chord lines.

B. Presentation of the different meshes

1. Global mesh parameters

Since the wingsail is composed of multiple elements, a structured mesh could not easily be created around it. Instead, the domain was partitioned into 7 concentric rings surrounding the wingsail. The innermost disk was meshed using an unstructured mesh, while the remaining subdomains were meshed with structured grids. To simplify identification and reference, each subdomain was labeled as follows:

- **Contours:** Numbered from 1 to 7, beginning at the outermost edge and moving inward to the innermost edge.
- **Faces:** Labeled from A to G, starting with the outermost face and progressing inward to the innermost face.

Each subdomain had a diameter a fraction of the domain outer diameter, δ , and was then assigned a mesh element size, for the edge and for the face, as a fraction of the the global element size S_G (Tab. II and III).

TABLE II: Mesh parameters for the edges of the mesh partition.

Edge	Ring diameter	Element Size (m)
1	δ	S_G
2	$0.4 \cdot \delta$	$0.75 \cdot S_G$
3	$0.3 \cdot \delta$	$0.5 \cdot S_G$
4	$0.25 \cdot \delta$	$0.4 \cdot S_G$
5	$0.2 \cdot \delta$	$0.3 \cdot S_G$
6	$0.15 \cdot \delta$	$0.2 \cdot S_G$
7	$0.1 \cdot \delta$	$0.1 \cdot S_G$

Additionally, since multiple wingsail configurations were tested, it was necessary to ensure convergence in each case. Consequently, four convergence analyses were performed for each extreme geometric configuration. Each convergence analysis is labeled with a group number and the corresponding geometric configuration is reported in Table. IV.

TABLE III: Mesh parameters for the faces of the mesh partition.

Face	Element Size (m)	Element Type	Structured?	Growth Rate
A	S_G	Quadrangle	Yes	1.2
B	$0.75 \cdot S_G$	Quadrangle	Yes	1.2
C	$0.5 \cdot S_G$	Quadrangle	Yes	1.2
D	$0.35 \cdot S_G$	Quadrangle	Yes	1.2
E	$0.2 \cdot S_G$	Quadrangle	Yes	1.2
F	$0.1 \cdot S_G$	Quadrangle	Yes	1.2
G	$0.07 \cdot S_G$	Triangle	No	1.1

TABLE IV: Slot width, camber, and angle of attack for each group in the convergence analysis.

Group	Slot Width s (% of C0 LE)	Base Camber γ ($^\circ$)	Angle of Attack α ($^\circ$)
0	0.5	15	0
1	0.5	35	15
2	5	15	0
3	5	35	15

2. Parameters for domain convergence

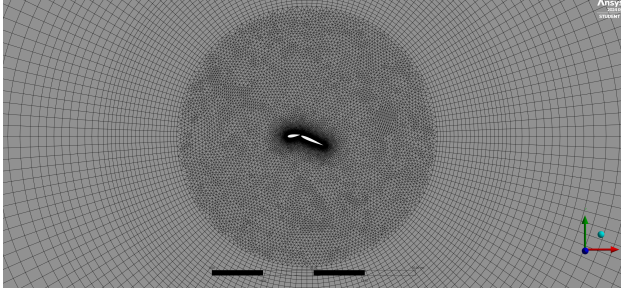
To asses the domain convergence, 7 simulations were carried out with increasing domain diameters, $\delta = 520, 560, 620, 640, 660, 720, 760$ m. The global mesh size used for this study was set to $S_G = 2$ m.

3. Local refinements and boundary layer

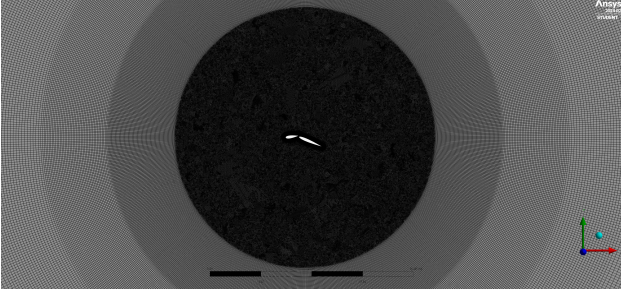
Each NACA contour (excluding the trailing edges) was assigned an edge size of $0.002 \cdot S_G$, while the truncated part of the trailing edges were discretized into 10 elements along their length. A sphere of influence with an element size of $0.002 \cdot S_G$ was also created between the two profiles.

Inflation layers were added to each profile to ensure the condition $y^+ < 1$ for the turbulence model. The first layer was set at 2×10^{-5} m, with 30 layers in total and a growth rate of 1.2.

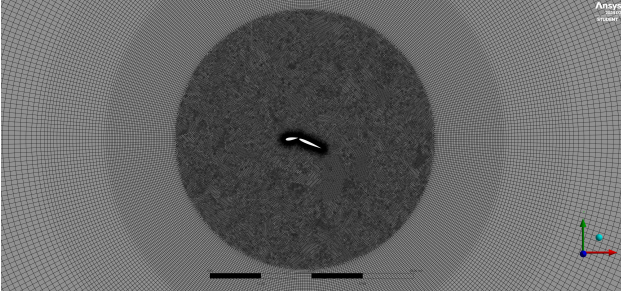
By changing the global element size S_G and the domain size δ , multiple meshes could easily be produced to verify convergence. Three meshes are presented in Figure 4 to visualize the effect of the global element size.



(a) Coarsest mesh, $S_G = 10$ m



(b) Finest mesh, $S_G = 2$ m



(c) Optimal mesh, $S_G = 4$ m

FIG. 4: Different mesh grid sizes used for the convergence analysis for the geometric configuration corresponding to group 3 (Tab. IV).

C. Mesh statistics for convergence analysis

To ensure a good convergence analysis, the global element size S_G and the domain size δ were varied in such a way that the number of elements increases sufficiently between each test case (Fig. 5). Furthermore, the mass imbalance is checked to be below $1 \times 10^{-3} \text{ kg s}^{-1}$ for every simulation except one, which is considered as acceptable (Fig. 6). Finally, the average mesh quality for the different simulations is shown to increase with domain diameter and global element size (Fig. 7). The mesh quality is defined as the ratio between the area of the element to the sum of the square of the edges lengths, multiplied by a co-

efficient. A value of 1 indicates a perfect square/triangle. Most elements decreasing the average quality are found in the boundary layer as they, purposefully, have a high aspect ratio.

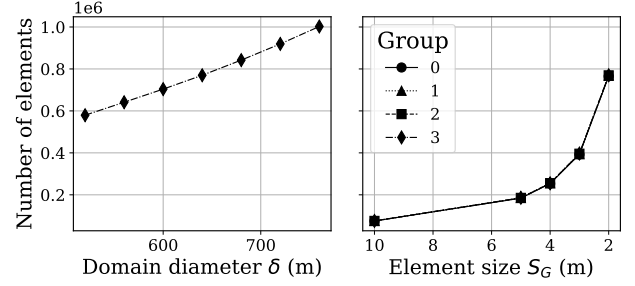


FIG. 5: Number of elements required for each convergence analysis.

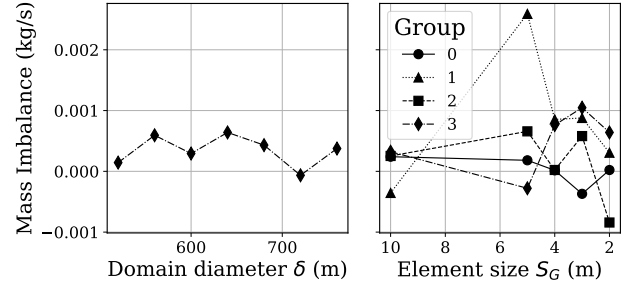


FIG. 6: Mass flux imbalance between inlet and outlet of the domain for different domain diameters and global element sizes

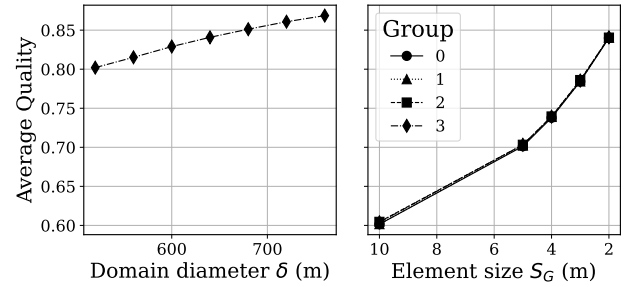


FIG. 7: Average mesh quality for the different domain diameters and global element sizes

D. Mesh convergence

Figures 8 and 9 present the convergence for both domain diameter and element size, respectively.

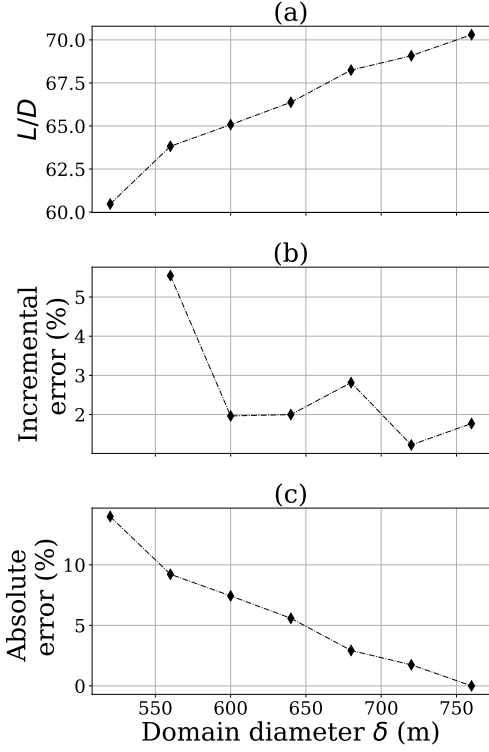


FIG. 8: Convergence analysis of the domain diameter for group 3 and a global mesh size $S_G = 2$ m.

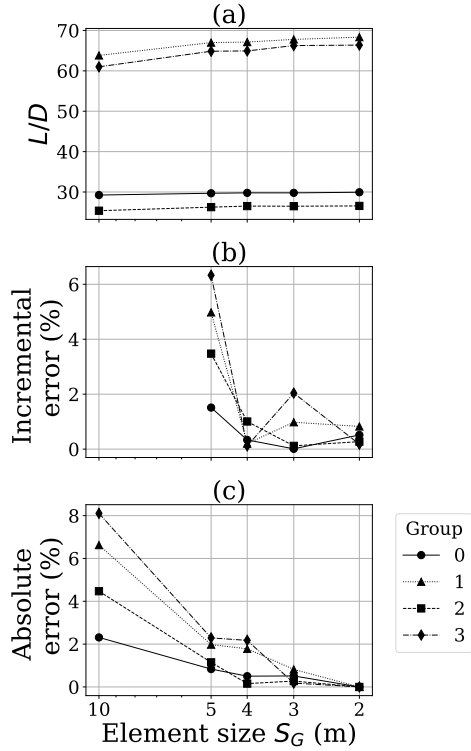


FIG. 9: Convergence analysis of the grid size.

E. Final mesh

Based on the criteria defined in section VI and the results presented in Figures 8 and 9, a domain diameter of $\delta = 640$ m and a global mesh size of $S_G = 4$ m were selected. The mesh, as detailed in Table III and II, consists of unstructured tetrahedral elements in the first region around the wingsail, transitioning to structured hexahedral elements for the subsequent rings. The final mesh is presented in different zoom levels in Figure 10.

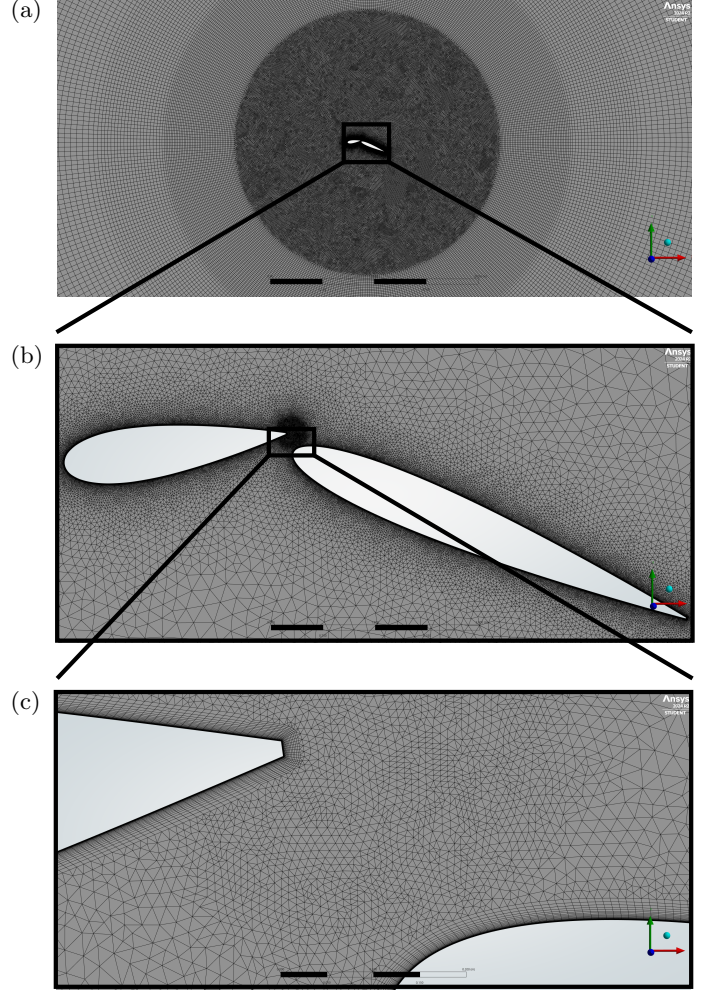


FIG. 10: Optimal mesh at different zoom levels, displaying the unstructured zone around the wingsail (a), the wingsail with the sphere of influence around the slot width (b), and the boundary layer with the local refinements on the NACA contour and trailing edge (c).

VII. RESULTS

The values of lift, drag, and their respective lift-to-drag ratio have been computed for each combination of the tested geometric parameters summarized in Table I. The

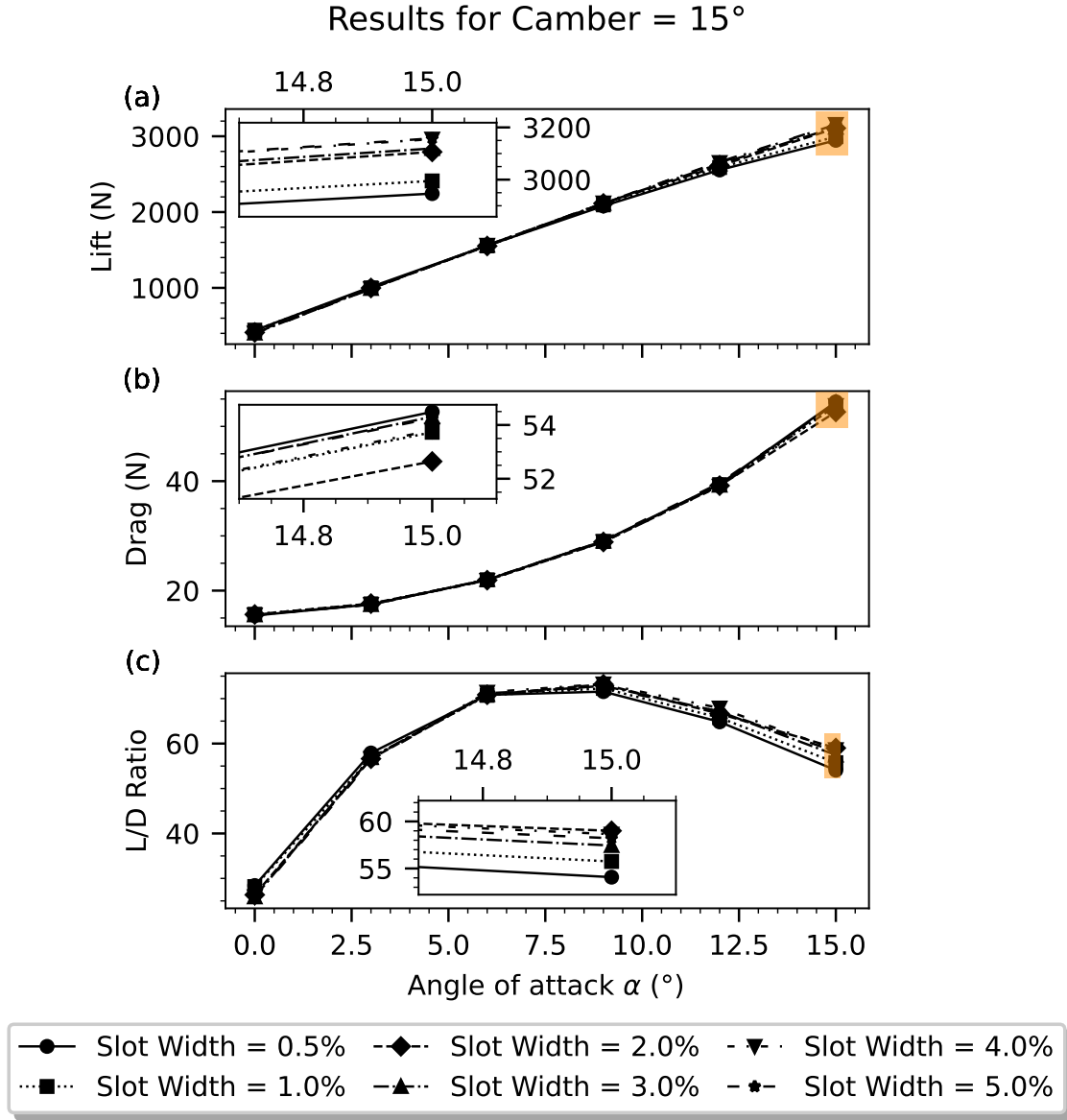


FIG. 11: Simulation results showing lift (a), drag (b), and lift-to-drag ratio (c) for various slot widths at a camber angle of 15° . A shaded orange region in each subplot highlights a magnified section for clearer visualization.

results for camber values of 15° and 35° are presented in figures 11 and 12 respectively. It can be observed that the lift and drag curves of each slot width overlap for most of the tested angles of attack in the 15° camber configuration. However, as the angle of attack increases, the effect of the slot width becomes clear as highlighted by the magnified region in figure 11. An increase in slot width produces more lift and less drag, which naturally increases the lift-to-drag ratio.

The effect of the slot width is however different for a higher camber. In the 35° configuration (Fig. 12) a decrease in lift can be observed throughout the angle of attack range for an increase in slot width, while drag is mostly unaffected. This produces a decrease in the lift-

to-drag ratio, particularly where the ratio is close maximal ($\alpha \approx 6^\circ$), as highlighted in the magnified areas.

To further investigate the effect of slot width on the maximum lift-to-drag ratio, the lift curve was linearly fitted and the drag curve was quadratically fitted. This approach enabled the determination of the optimal lift-to-drag ratio for each configuration, along with the corresponding optimal angle of attack, using all of the data points and ensuring better accuracy (Fig. 13). The resulting curves as a function of slot width indicate only a small effect on the optimal angle of attack, while the maximum lift-to-drag ratio varies by several percent, positively and negatively for the 15° and 35° configurations, respectively.

Results for Camber = 35°

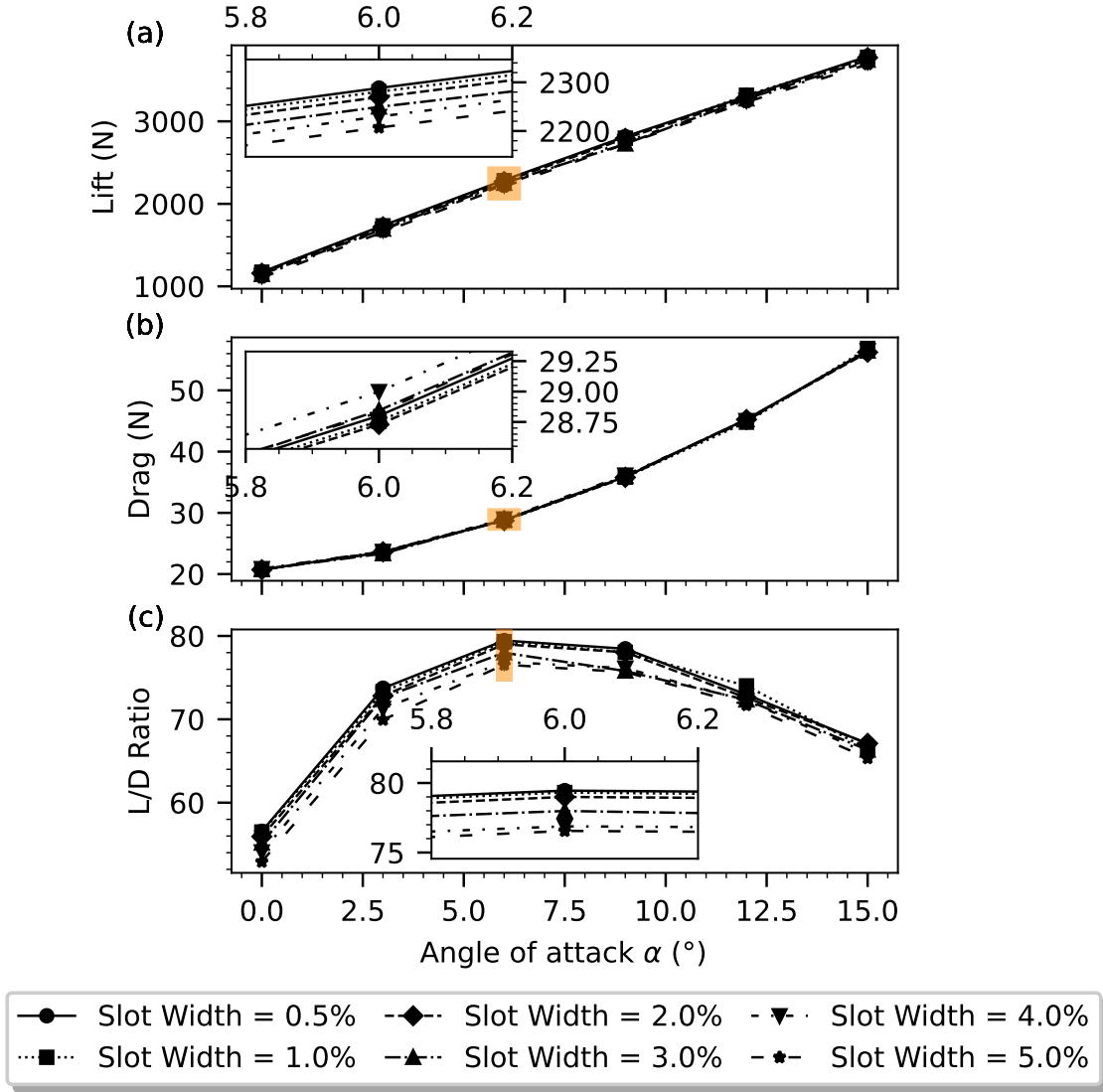


FIG. 12: Simulation results showing lift (a), drag (b), and lift-to-drag ratio (c) for various slot widths at a camber angle of 35°. A shaded orange region in each subplot highlights a magnified section for clearer visualization.

To gain a better understanding of the reason for the loss in performances for a camber of 35°, the pressure coefficient and skin friction coefficient have been plotted for the two extreme slot width values, namely 0.5 % and 5 % (Fig. 14). The results suggest that the pressure difference is decreased throughout the LE, but that the peak on the suction side of the TE is increased for a larger slot width. A similar trend is observed for the friction coefficient.

To visualize the flow field in the gap between the LE and TE, a contour plot of the velocity magnitude and gauge pressure is presented (Fig. 15). The decrease in pressure on the suction side of the nose of the TE for an increased slot width is confirmed and can also be observed from the increase in velocity in that region.

VIII. DISCUSSION

The results presented in the previous section give a mixed interpretation of the effect of the slot width on the wingsail performance. On the one hand, the lift is increased with a decrease in drag for a high angle of attack and a low camber configuration, on the other hand, the high camber configuration shows a decrease in lift throughout the angle of attack range, although the performance reduction is smaller for high angles of attack, suggesting that the optimal slot width is larger for increased angles of attack.

The ideal trimming of the sail being a complicated function of sailing conditions and boat geometry, it is not

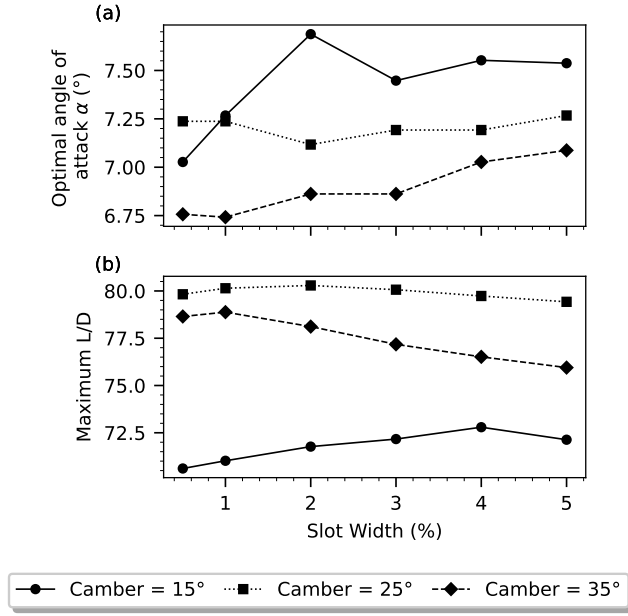


FIG. 13: Simulation results showing the angle of attack that maximizes the lift-to-drag ratio (a), and corresponding lift-to-drag value (b), as a function of slot width, for camber angles of 15°, 25° and 35°.

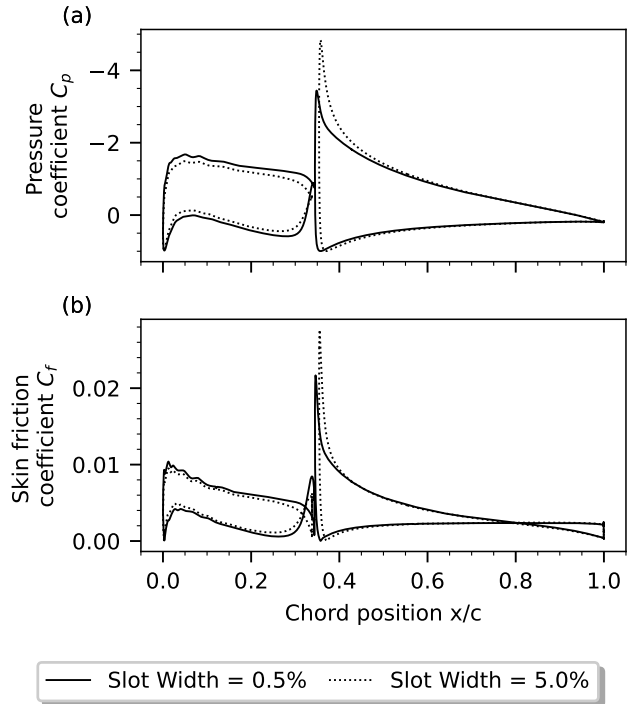


FIG. 14: Pressure coefficient (a) and skin friction coefficient (b) as a function of normalized chord position for an angle of attack of 6°, a camber of 35° and slot widths of 0.5% and 5%.

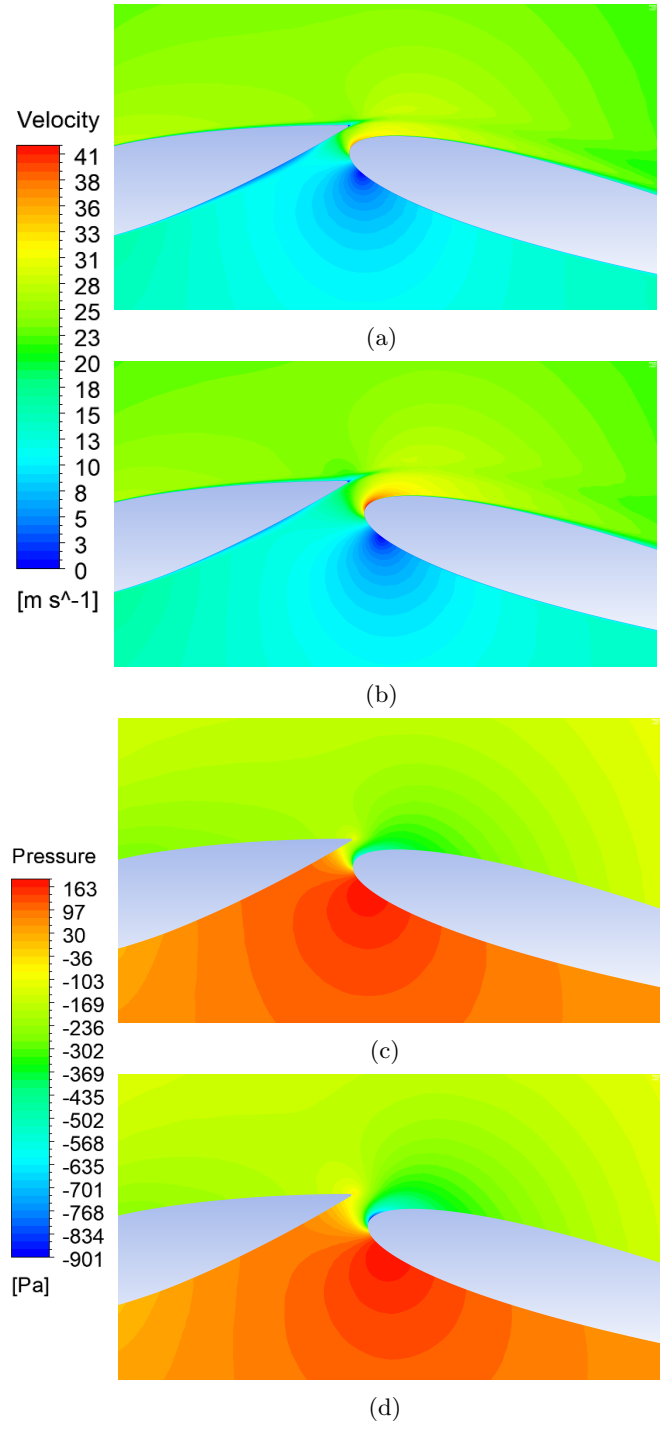


FIG. 15: Contour plots around the slot, of velocity magnitude (a, b) and pressure (c, d) for a camber of 35° and an angle of attack of 6°. Results are shown for two geometric configurations: slot widths of 0.5% (a, c) and 5% (b, d).

possible to decouple the hydrodynamic and aerodynamic problems [4]. Thus, the angle of attack maximizing the boat speed is not necessarily the one maximizing the lift-to-drag ratio [4], nor is it the one maximizing lift, but actually lies between the two. Predicting this ideal angle of attack becomes important in deciding which slot width yields better performances.

In this study, the prediction of the wingsail performances was limited to relatively low angles of attack because of the choice of performing a RANS simulation. It has been shown that RANS simulations using the $k-\omega$ SST turbulence model do not give reliable stall characteristics, even for light stall [5]. Thus, it was decided to avoid simulating those conditions to avoid misinterpreting the resulting data. Nevertheless, considering the important role of angle of attack on the slot width performances and considering that the potential of the leakage flow helping delay separation and promoting flow reattachment is more relevant at higher angles of attack, a future study properly modeling stall would be relevant.

IX. CONCLUSION

In conclusion, this study revealed the complex role of the gap between the two elements of the wingsail on its aerodynamic performances. It has been highlighted that an increased slot width, from 0.5% to 5% of the percentage relative to the chord length of the LE, measured at chord number 0, can yield either positive or negative impacts on the sail characteristics of the order of several percent. The precise effect of the gap and its optimal size are dependent on the sailing conditions and future work is required to properly analyze higher angles of attack and stall characteristics.

Appendix: y^+ of the wingsail

As shown in Figure 16, the y^+ are globally below 1, except at the nose of the TE, where a peak reaching 1.4 is observed, which was deemed acceptable.

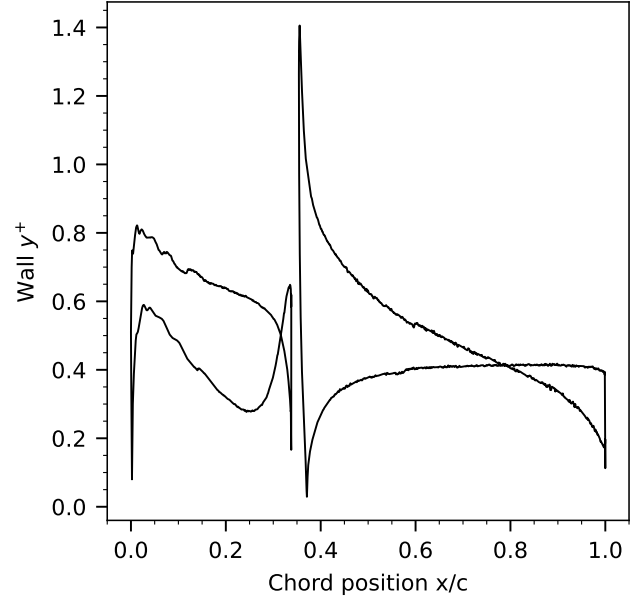


FIG. 16: y^+ of both NACA profiles of the optimal mesh in the configuration with a camber of 35° , an angle of attack of 6° and a slot width of 5%.

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